Towards Trustworthy and Reliable Deployment of Adversarial Attack Detection for Brain MRI Classification

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Abstract. Adversarial attack detectors are commonly reported using ROC-AUC or detection rates at a false-positive rate (FPR) calibrated on training-side data. On a four-class brain-tumor MRI task, we show that this convention overstates deployed performance: thresholds targeting a 10% FPR achieve 13.56% and 15.44% on the test collection, while ROC-AUC remains high. Recalibration using 400 clean images from the deployment collection reduces the evaluation FPR to 10.75%, without requiring labelled attacks. At this fixed operating point, an FGSM-trained detector also detects unseen PGD attacks within two percentage points of its FGSM performance on identical images. The reported results were reproduced independently in two Google Colab T4 runs.

Keywords: Adversarial examples · Attack detection · Medical imaging · Calibration · False positive rate.

1 Introduction

Deep networks for medical image classification are easy to fool with imperceptible perturbations [2], and the incentives to fool them are concrete in medicine: insurance and billing fraud, manipulation of imaging endpoints in trials, and sabotage of automated triage have all been articulated as realistic motivations [1]. A body of work has answered with detectors that flag adversarial inputs. For medical imaging the literature's headline is optimistic: adversarial perturbations distort feature statistics so strongly that simple detectors reach over 98% AUC [3]. Yet a detector is deployed at a *threshold*, not at an AUC, and in a hospital the cost of a false alarm (a repeated scan, a delayed diagnosis) makes the achieved false-positive rate a condition of use rather than an implementation detail.

This paper asks what happens to the operating point between the lab and deployment, using a brain-tumor MRI classifier as the test bed. Our contributions:

1. **A measured failure of threshold transfer.** On a standard Kaggle brain-tumor dataset whose `Training/` and `Testing/` folders are distinct collections, a detector threshold selected for FPR $\leq$ 10% on validation data achieves 13.6% on test; a leakage-free calibration split makes it 15.4% (Table 2). The drift is confined to the upper tail of the clean-score distribution, exactly where thresholds operate, while ROC-AUC remains high — so AUC-only reporting cannot reveal the operating-point failure.
2. **A measured remedy.** Calibrating on deployment-side data is standard practice; what we contribute is the measurement that, on this task, 400 *clean* images from the deployment collection reduce the evaluated FPR to 10.75% (a +0.75-point error, compared with +5.44 points for leakage-free training-side calibration) — no attack examples are needed, matching what a hospital actually has.
3. **Honest re-pricing, and cross-attack transfer at the pinned point.** At the deployment-calibrated 10.75% FPR, the detection rate for successful FGSM attacks is 36.0% ($\varepsilon = 0.01$), 73.4% (0.05), 86.1% (0.1), and 96.5–98.5% ($\varepsilon \geq 0.25$). Trained on FGSM only, the detector detects successful PGD attacks (10 and 40 steps, random start) within ±1–3 points of these numbers on the *same* images (intersection protocol, Table 5). All paired differences are below two percentage points.
4. **A strict-dominance observation.** At every ε tested, every image broken by FGSM is also broken by PGD ($n_{\text{FGSM only}} = 0$ in all 12 pairings) — direct evidence that FGSM-only robustness evaluation is optimistic.

We deliberately label only *successful* attacks (a correct diagnosis turned incorrect) as positives: unsuccessful perturbations are noise for a clinical threat model. All experiments run under a pipeline in which every model artifact is SHA-256-chained and every downstream stage verifies hashes and clean accuracy before computing.

2 Related Work

Consistency-based detection. Feature squeezing [9] compares predictions before and after input squeezing (bit-depth reduction, blur) and thresholds the L_1 difference, selecting the threshold on clean data for a target FPR. Our detector learns from a richer consistency representation (penultimate features, class probabilities, blur-consistency summaries, confidence/margin/entropy) but inherits the clean-calibrated-threshold idea — and measures what [9] did not: whether that threshold survives a collection shift.

Detection in medical imaging. Ma et al. [3] established that medical adversarial examples are easy to detect (>98% AUC on X-ray, fundoscopy, dermoscopy). Our AUCs agree; our point is that the AUC-to-operating-point gap is where deployment fails. A DFT-based detector has been proposed for MRI Alzheimer's classification [5]; it labels all adversarial inputs positive and does not examine FPR transfer.

Defense on the same task. A recent multi-layered defense for brain tumor classification [10] combines adversarial training with feature squeezing on the same dataset family — complementary robustness, not detection; no detector, FPR, or calibration is involved.

Cross-attack generalization. That detectors trained on one gradient attack generalize to others is documented in the natural-image literature; AED-PADA [7] improves it further via multi-source domain adaptation. We therefore do not claim cross-attack transfer as novel; our contribution is measuring it *at a pinned deployment operating point* and on an intersection of success sets, which removes the changing-denominator confound.

3 Experimental Setup

Task and data. Four-class brain-tumor MRI classification (glioma, meningioma, no-tumor, pituitary), 5,600 training and 1,600 test images (balanced, 224×224, pixel range 0–255) [6]. The training folder is split 80/20 into 4,480 train / 1,120 validation source images with verified zero overlap.

Dataset audit. Because our central claim concerns a shift between the dataset's two folders, we audited all 7,200 images. Three findings (Table 1). (i) *Test–train duplication:* 100 of the 400 meningioma test images are pixel-identical, re-encoded duplicates of Training images (verified by z-normalised downsampled matching with full-resolution confirmation; 64-bit perceptual hashes produce false alarms on MRI slices and were not used). Excluding them *raises* clean accuracy (81.9% $\rightarrow$ 83.6% overall; meningioma 64.3% $\rightarrow$ 67.0%), so the contamination does not inflate our results — we disclose it rather than correct for it. (ii) *Manufactured balance:* the meningioma class in both folders is padded with synthetic augmentations (100 train / 103 test) to reach round per-class counts; these synthetic images contain no duplicates. (iii) *A banner-marked subpopulation:* many images carry burned-in text or colour banners from their web sources. Their prevalence differs sharply between folders for glioma (1.2% of Training vs. 13.3% of Testing), and the classifier collapses on exactly that subpopulation: 15.1% accuracy on banner-flagged glioma test images vs. 74.4% on unflagged ones — a reminder that a deployed model can fail almost completely on an identifiable acquisition subpopulation while aggregate accuracy looks acceptable. The collection shift behind our FPR finding is thus concrete and visible in the pixels, not only in detector score statistics. A shortcut-learning hypothesis for the no-tumor class (61% banner prevalence, the classifier's best class) was tested and refuted: flagged no-tumor images are classified slightly *worse* (-1.6 points).

Classifier. ImageNet-pretrained MobileNetV2 [8], frozen backbone, light augmentation, trained head (Adam 10^{-3}, ≤ 10 epochs, early stopping); clean test accuracy 81.94% (1,311/1,600). This classifier is a testbed for the calibration question, not a clinical tool — its accuracy, particularly on meningioma, is far below any deployable bar. Training is fully seeded with op determinism, and the canonical model file is never overwritten — downstream stages refuse

Table 1. Dataset audit: banner-flagged image prevalence and its effect on clean test accuracy, per class.

Class	Flagged (Train)	Flagged (Test)	Acc. flagged	Acc. unflagged
Glioma	1.2%	13.3%	15.1%	74.4%
Meningioma	9.9%	21.5%	66.3%	63.7%
No tumor	62.9%	60.5%	97.1%	98.7%
Pituitary	15.7%	16.0%	98.4%	99.1%

to run if its hash changes. The reported pipeline was run independently by both authors on Google Colab Tesla T4 GPUs with TensorFlow 2.20.0 and seed 42; both runs produced the same evaluation metrics.

Attacks. Untargeted white-box FGSM [2] on the 0–255 input scale at $\varepsilon \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0\}$[3]; PGD [4] with random start, step $\alpha = \varepsilon/4$, $K \in \{10, 40\}$, with per-batch assertions keeping every perturbation inside the ε-ball. FGSM at $\varepsilon = 1$ already yields 95.0% attack success on initially-correct images; PGD-40 reaches 100.0% (979/979 on the evaluation split) — the undefended classifier offers no robustness at that budget.

Detector. A two-layer MLP (256/64, dropout) on a consistency feature vector extracted from the *frozen* classifier: penultimate GAP features, class probabilities before and after a light average-pool blur, their differences, and confidence/margin/entropy summaries. Positives are successful attacks only, at training $\varepsilon \in \{0.01, 0.1, 0.5\}$; class weights balance the loss. Model selection uses a 560-image validation half (val_A); the other 560 images (val_B) are reserved for threshold calibration and touch neither training nor selection.

Evaluation protocol. The 1,600 test images are split (seeded) into 400 calibration / 1,200 evaluation images. Every reported test number comes from the 1,200 evaluation images at one fixed threshold; the test set is never used for tuning. ε values 0.05, 0.25, 1.0 are unseen by the detector.

Pipeline integrity. Security evaluations are unusually sensitive to silent artifact drift: if the classifier is ever retrained between the detector-fitting and final-evaluation stages, the test tables mix incompatible eras without any error being raised. Our pipeline therefore records the SHA-256 of every trained model in a manifest together with its clean test accuracy, and every downstream stage re-verifies both before computing anything, failing closed on mismatch. Training is fully seeded with deterministic ops, so two independent Colab T4 executions reproduced the reported evaluation metrics. Serialized model hashes may differ because of file-level metadata, so we claim reproducibility of evaluated results rather than byte-identical model archives. The design was motivated by exactly such a silent-retrain incident in an earlier iteration of this study.

[3] On the usual $[0, 1]$ convention, $\varepsilon = 1$ here is $1/255 \approx 0.0039$ — small budgets by natural-image standards, which makes the near-total attack success at $\varepsilon \geq 0.5$ the more striking.

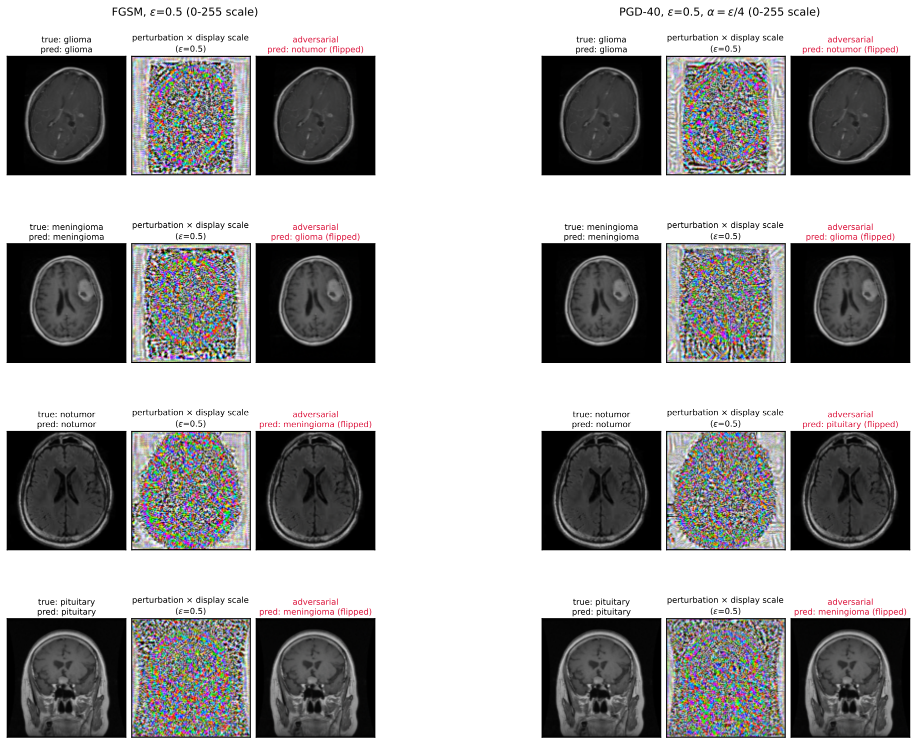

Fig. 1. One example per class at $\varepsilon = 0.5$ under FGSM (left) and PGD-40 (right): original, perturbation (amplified for display), and adversarial image with the flipped prediction. The perturbation is imperceptible at native contrast, yet every diagnosis changes. Exemplars were screened to exclude banner-marked, synthetic, and duplicated images (Sect. 3, dataset audit).

4 The Operating Point Does Not Transfer

Three calibration strategies against one budget (FPR $\leq$ 10%):

Removing selection leakage (row 2) does not help: the test FPR increases from the 8.93% calibration value to 15.44%. This indicates a shift in the upper tail of the clean-score distribution, exactly where a fixed threshold operates. ROC-AUC remains high under the same shift, which is why AUC-only reporting cannot reveal the problem. A margin variant (calibrating to 7.5% for headroom) also misses its target, reaching 14.25% on the full test set.

Calibrating on 400 clean images from the deployment collection lands the threshold at 0.321, reducing the evaluation FPR to 10.75%, and requires no labelled attacks: a realistic protocol for a hospital that has clean scans but no attack examples. Deployment-side calibration is standard practice in principle; the point here is empirical — how badly the training-side shortcut misses, and how little data suffices. Two scoping remarks: the 10% budget is illustrative (the

Table 2. Claimed vs. achieved false-positive rate under three calibration strategies.

Calibration data	Claimed FPR	Achieved test FPR	Error
Validation, 1,120 imgs (also used for selection)	$\leq$ 10%	13.56%	+3.56
val_B, 560 imgs (leakage-free)	$\leq$ 10%	15.44%	+5.44
400 clean deployment-collection imgs	$\leq$ 10%	**10.75%**	**+0.75**

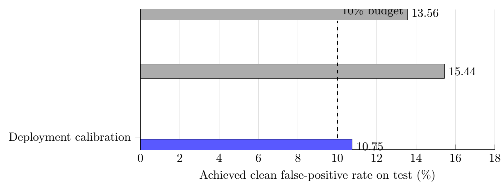

Fig. 2. Achieved clean false-positive rate on the test collection under the three calibration strategies (budget: 10%). Deployment-side calibration comes closest to the target.

protocol pins whatever budget a site chooses; the 7.5% margin variant behaves identically), and the calibration set is assumed attack-free — an adversary who can poison those 400 images moves the threshold at will, so their collection must be controlled and auditable.

Cost of honesty. At the pinned threshold, low-ε detection is lower than the mis-calibrated numbers suggested: 73.4% instead of 83.2% at $\varepsilon = 0.05$ (the former on the 1,200-image evaluation subset, the latter on the full test set at its actual 15.4% FPR — the comparison mixes evaluation sets, but both the direction and the size survive the difference). The earlier figure was bought with about 54% more false alarms than the 10% target. Table 3 gives the complete re-priced results. At $\varepsilon = 0.01$, PR-AUC is 0.075 (25 positives vs. 1,200 cleans): tiny attacks remain essentially undetectable, and ROC-AUC (0.887) flatters this regime badly — we quote PR-AUC alongside ROC-AUC at all small ε.

Table 3. Complete FGSM results on the 1,200 evaluation images at the deployment-calibrated operating point (clean FPR 10.75%). "Margin" is the FPR$\leq$7.5% calibration variant (clean FPR 8.25%). ε values marked † were never seen in detector training. The $\varepsilon = 0.01$ row lies below 8-bit input quantization (Sect. 6) and is reported for completeness only.

ε	Successful attacks	Detection	Detection (margin)	ROC-AUC	PR-AUC
0.01	25	36.0%	28.0%	0.876	0.075
0.05†	139	73.4%	63.3%	0.918	0.496
0.10	310	86.1%	83.2%	0.956	0.833
0.25†	665	96.5%	95.8%	0.985	0.973
0.50	865	98.3%	97.9%	0.993	0.991
1.00†	929	98.5%	98.3%	0.995	0.994

5 Cross-Attack Transfer at the Pinned Operating Point

Trained on FGSM only, evaluated against PGD it has never seen, at the same fixed threshold. Table 4 reports the raw PGD results: PGD is the stronger attack at every ε (at $\varepsilon = 1$, PGD-40 breaks all 979 initially-correct evaluation images), yet detection rates and AUCs remain at FGSM levels. Because PGD breaks more images than FGSM, raw detection rates have different denominators; to remove that confound we compare on the intersection of success sets (Table 5).

Table 4. PGD attacks against the FGSM-trained detector at the same deployment threshold (clean FPR 10.75%). Attack success is measured on the 979 initially-correct evaluation images; PR-AUC (not shown) tracks FGSM's (0.08 at $\varepsilon = 0.01$, ≥ 0.56 from 0.05, ≥ 0.98 from 0.25). The $\varepsilon = 0.01$ rows lie below 8-bit input quantization and are reported for completeness only.

	PGD-10			PGD-40		
ε	Succ.	Det.	ROC-AUC	Succ.	Det.	ROC-AUC
0.01	2.6%	36.0%	0.876	2.7%	34.6%	0.877
0.05	16.9%	74.5%	0.922	17.4%	74.1%	0.923
0.10	40.6%	88.2%	0.960	41.9%	88.5%	0.960
0.25	87.1%	97.0%	0.988	89.5%	97.3%	0.989
0.50	98.6%	98.1%	0.994	99.4%	98.4%	0.995
1.00	99.9%	99.1%	0.996	100.0%	99.0%	0.996

Table 5. Detection rates of successful attacks on the *intersection* of FGSM and PGD success sets (identical images), at the deployment-calibrated threshold (clean FPR 10.75%). The $\varepsilon = 0.01$ row lies below 8-bit input quantization and is reported for completeness only.

ε	n (intersection)	FGSM	PGD-10	PGD-40
0.01	25	36.0%	36.0%	36.0%
0.05	139	73.4%	74.8%	74.1%
0.10	310	86.1%	87.7%	88.1%
0.25	665	96.5%	96.5%	96.7%
0.50	865	98.3%	97.9%	98.3%
1.00	929	98.5%	99.0%	98.9%

On resolution: at these sample sizes the per-row 95% binomial half-width is $\approx \pm 1$ point at $p \approx 0.98$ ($n = 865$–929) and ± 7 points at $n = 139$, so individual row differences sit at the noise floor; the evidence is consistency of sign across ε and both K on paired (identical-image) comparisons. A paired McNemar-style test from the persisted per-image scores is future work.

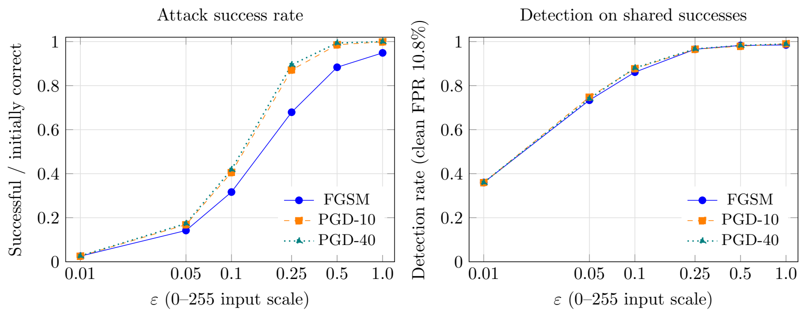

Fig. 3. Left: attack success rate on initially-correct evaluation images. Right: detection rate of successful attacks on the intersection of success sets, at the deployment-calibrated threshold (clean FPR 10.8%). The detector was trained on FGSM only.

Two observations. **(1) Transfer holds.** On identical successful images, every PGD detection rate is within two percentage points of its FGSM counterpart; ROC-AUC is also nearly identical throughout (0.876–0.996). The rerun does not support a consistent crossover in detectability, so we make no claim about the sign of these small paired differences. **(2) Strict dominance.** $n_{\text{FGSM only}} = 0$ in every pairing: no image at any ε is broken by FGSM but not by PGD, so reporting robustness against FGSM alone is provably optimistic on this task. The images only PGD can break at $\varepsilon \geq 0.25$ are detected at 98–100% — resisting a one-shot attack forces a perturbation the detector sees clearly.

Robustness to the dataset defects. Excluding the synthetic meningioma images, the duplicated ones, or both changes every $\varepsilon \geq 0.1$ intersection entry by about one point or less; the $\varepsilon = 0.01$ row swings ($54\% \rightarrow 33\%$) purely from its tiny $n \leq 25$. The dataset-defect exclusion analysis was produced with the original model and is retained as supplementary evidence; it must be rerun with the final Colab model before its exact FPR values are treated as part of the primary result set.

6 Limitations

$\varepsilon = 0.01$ on the 0–255 scale is below one 8-bit quantization level; FGSM and PGD-10 are nearly identical there, its success counts are tiny ($n \leq 25$), and its detection rate is unstable under the dataset-exclusion scenarios — the row is reported for completeness only. All attacks target the classifier; none optimize to evade the detector — a Carlini-style adaptive evaluation is the necessary next step before any strong defensive claim. The FPR-transfer finding is measured on one (widely used) dataset whose training/testing split happens to be a collection split, with one backbone; the audit in Sect. 3 shows this dataset also carries verbatim test–train duplicates and synthetic padding, which we disclose and show do not drive our conclusions, but breadth across cleaner modalities is needed before generalising. The evaluation set is 75% of the original test set (400 images are spent on calibration). Per-image scores and masks are persisted for future paired analyses.

7 Conclusion

On a real medical-imaging dataset, the gap between a detector's AUC and its deployed operating point is where the security claim quietly breaks: every training-side threshold missed its FPR budget, while 400 clean deployment images — no attacks required — brought it close to the target. We recommend that detection papers (i) report the achieved FPR on the evaluation distribution next to every detection rate, (ii) treat threshold-free metrics as insufficient for deployment claims, and (iii) compare attack families on intersections of success sets. Under this stricter accounting our FGSM-trained consistency detector remains effective against the non-adaptive attacks evaluated here: at least 86.1% detection of successful attacks at $\varepsilon \geq 0.1$ at an achieved clean FPR of 10.75%, with attack-family differences below two percentage points on identical images. Whether it survives an adversary that optimizes against the detector itself is an open question this paper does not answer.

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